DEPARTMENT OF PHYSICS
INDIAN INSTITUTE OF TECHNOLOGY, MADRAS

PH1020 Physics II

Syllabus 2025

Learning objectives:

To enable the students to:

1) describe electromagnetic phenomena using the language of vector calculus.

2) determine electric and magnetic fields arising from simple configurations of static charges as well as steady currents in vacuum and in matter.

3) solve problems involving the propagation of electromagnetic waves in vacuum and in matter.

Text Book:

Introduction to Electrodynamics
by *David J. Griffiths*
4th Edition (South Asia Edition)

Detailed Syllabus with section numbers from Griffith 4rd Edition:

Unit 1: Electrostatics & Electric Fields in Matter

[2.1] The Electric Field, [2.1.2] Coulomb's Law, [2.1.3] The Electric Field, [2.1.4] Continuous Charge Distributions, [2.2] Divergence and Curl of Electrostatic Fields, [2.2.1] Field Lines, Flux, and Gauss's Law, [2.2.3] Applications of Gauss's Law, [1.5] The Dirac Delta Function, [2.3] Electric Potential, [2.3.4] The Potential of a Localized Charge, Distribution, [2.3.5] Boundary Conditions, [2.4] Work and Energy in Electrostatics, [2.4.1] The Work It Takes to Move a Charge, [2.4.2] The Energy of a Point Charge Distribution, [2.4.3] The Energy of a Continuous Charge Distribution, [2.5] Conductors, [2.5.1] Basic Properties, [2.5.2] Induced Charges, [2.5.3] Surface Charge and the Force on a Conductor, [2.5.4] Capacitors, [2.3.3] Poisson's Equation and Laplace's Equation, [3.1.2] Laplace's Equation in One Dimension, [3.4] Multipole Expansion, [3.4.2] The Monopole and Dipole Terms, [LN1]*Quadrupole moment tensor*, [3.4.4] The Electric Field of a Dipole, [4.1.3] Force and torque on a dipole due to an external static electric field, [LN2] *Energy of an ideal dipole in an electric field, The interaction energy between two dipoles.*

Electric Fields in Matter: [4.1] Polarization, [4.1.1] Dielectrics, [4.1.2] Induced Dipoles, [4.2] The Field of a Polarized Object, [4.2.1] Bound Charges, [4.3] The Electric Displacement, [4.3.1] Gauss's Law in the Presence of Dielectrics, [4.3.3] Boundary Conditions, [4.4] Linear Dielectrics, [4.4.1] Susceptibility, Permittivity, Dielectric Constant, [4.4.3] Energy in Dielectric Systems

Unit 2: Magnetostatics & Magnetic Fields in Matter

[5.1] The Lorentz Force Law, [5.1.1] Magnetic Fields, [5.1.2] Magnetic Forces, [5.1.3] Currents, [5.2] The Biot-Savart Law, [5.2.1] Steady Currents, [5.2.2] The Magnetic Field of a Steady Current, [5.3.1] Straight-Line Currents, [5.3.2] The Divergence and Curl of B, [5.3.3] Ampère's Law, [5.4] Magnetic Vector Potential, [5.4.1] The Vector Potential, [5.4.2] Boundary Conditions, [5.4.3] Multipole Expansion of the Vector Potential.

Magnetic Fields in Matter: [6.1] Magnetization, [6.1.1] Diamagnets, Paramagnets, Ferromagnets, [6.1.2] Torques and Forces on Magnetic Dipoles, [6.2] The Field of a Magnetized Object, [6.2.1] Bound Currents, [6.3] The Auxiliary Field H, [6.3.1] Ampère's Law in Magnetized Materials, [6.3.3] Boundary Conditions, [6.4] Linear and Nonlinear Media, [6.4.1] Magnetic Susceptibility and Permeability.

Unit 3: Electrodynamics & Electromagnetic Waves

[7.1] Electromotive Force, [7.1.1] Ohm's Law, [7.1.2] Electromotive Force, [7.1.3] Motional emf, [7.2] Electromagnetic Induction, [7.2.1] Faraday's Law, [7.2.2] The Induced Electric Field, [7.2.3] Inductance, [7.2.4] Energy in Magnetic Fields, [7.3.1] Electrodynamics Before Maxwell, [7.3.2] How Maxwell Fixed Ampère's Law [7.3.3] Maxwell's Equations, [7.3.5] Maxwell's Equations in Matter, [7.3.6 Boundary Conditions, [10.1, LN3] *Gauge invariance-Coulomb and Lorentz gauge*, [8.1.1] The Continuity Equation, [8.1.2] Poynting's Theorem, [8.2.4] Angular Momentum.

Electromagnetic Waves: [9.2] Electromagnetic Waves in Vacuum, [9.2.1] The Wave Equation for E and B, [9.2.2] Monochromatic Plane Waves, [9.2.3] Energy and Momentum in Electromagnetic Waves, [9.3] Electromagnetic Waves in Matter, [9.3.1] Propagation in Linear Media, [9.3.2] Reflection and Transmission at Normal Incidence, [9.3.3] Reflection and Transmission at Oblique Incidence (Law of reflection, law of refraction—Snell's law, Fresnel's equations, Brewster's angle)

**LN = Lecture Notes*